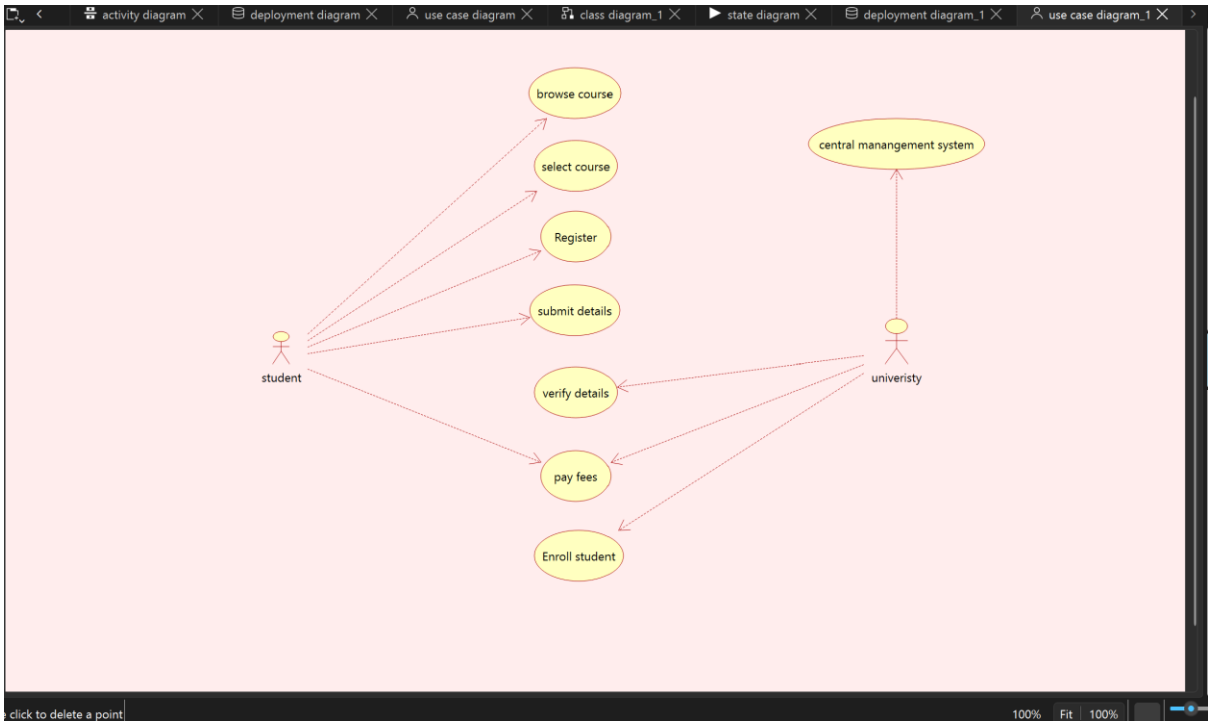
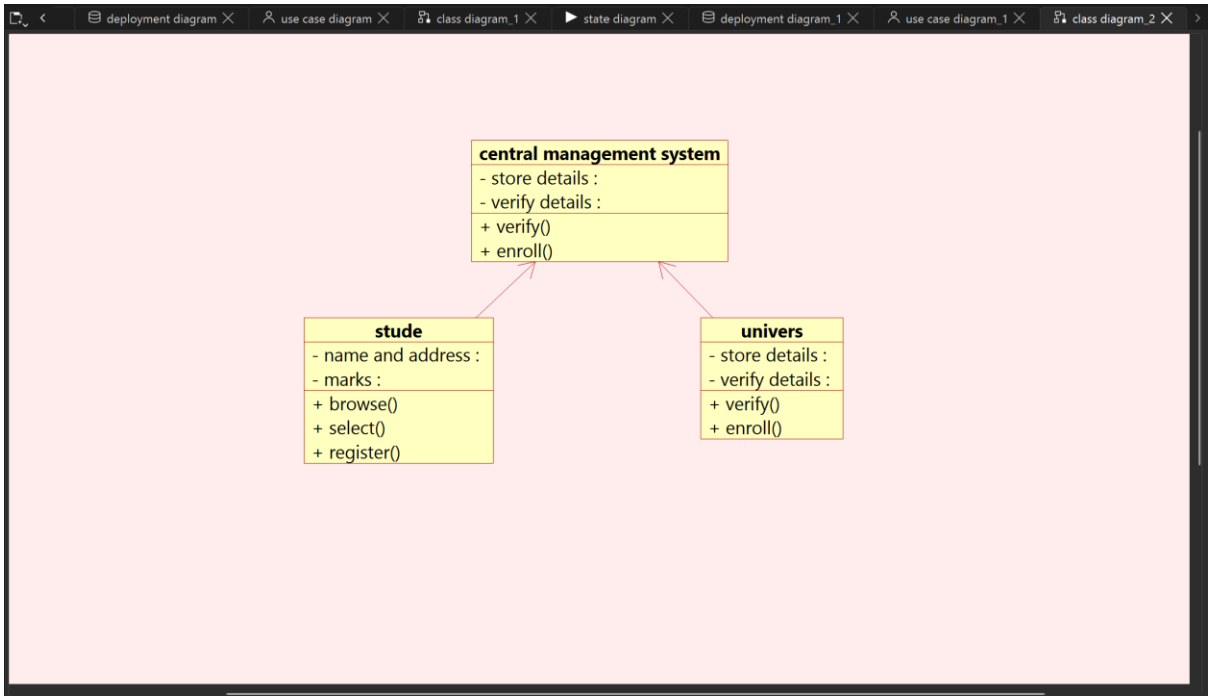
Ex No: 5 – ONLINE COURSE RESERVATION SYSTEM

NAME: HEMALAKSHMI H (192571049)

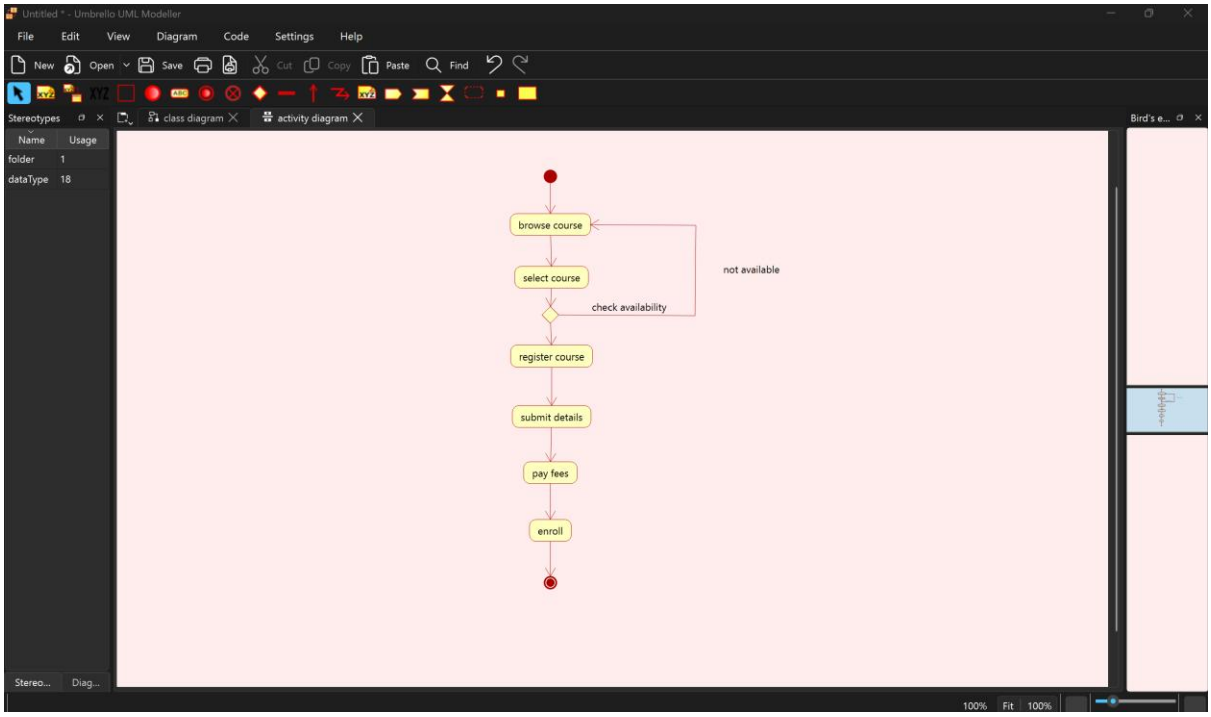
USE CASE DIAGRAM:



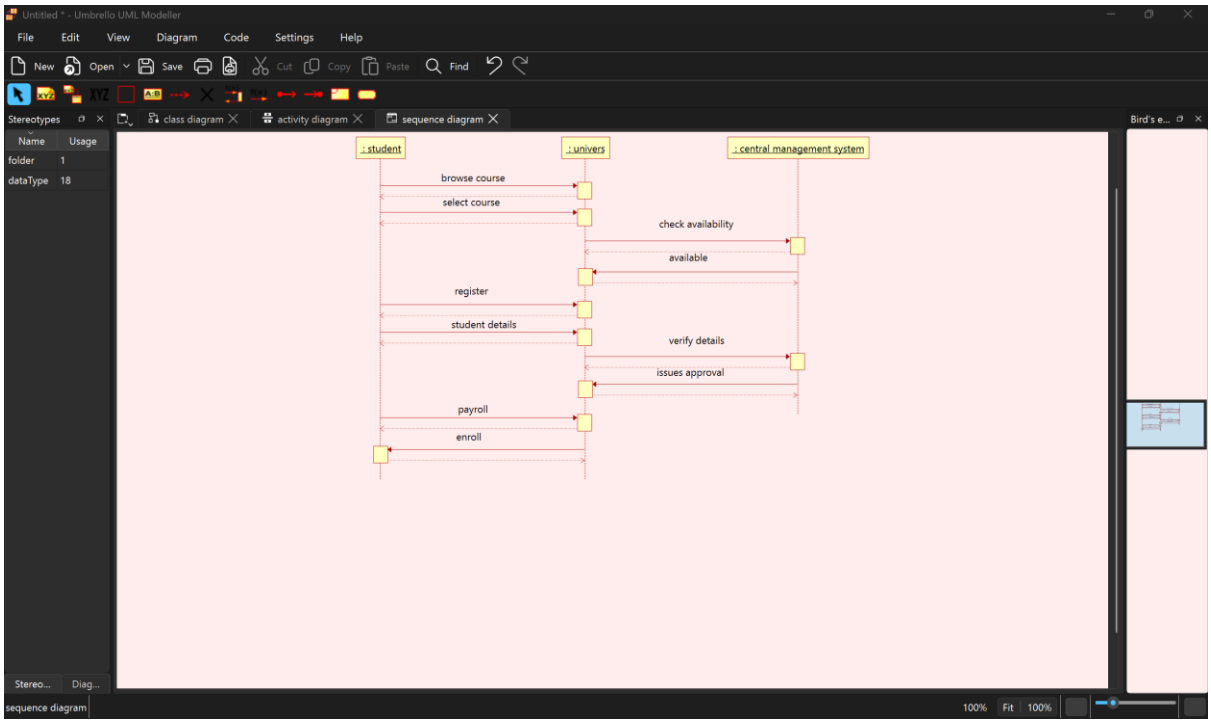
CLASS DIAGRAM:



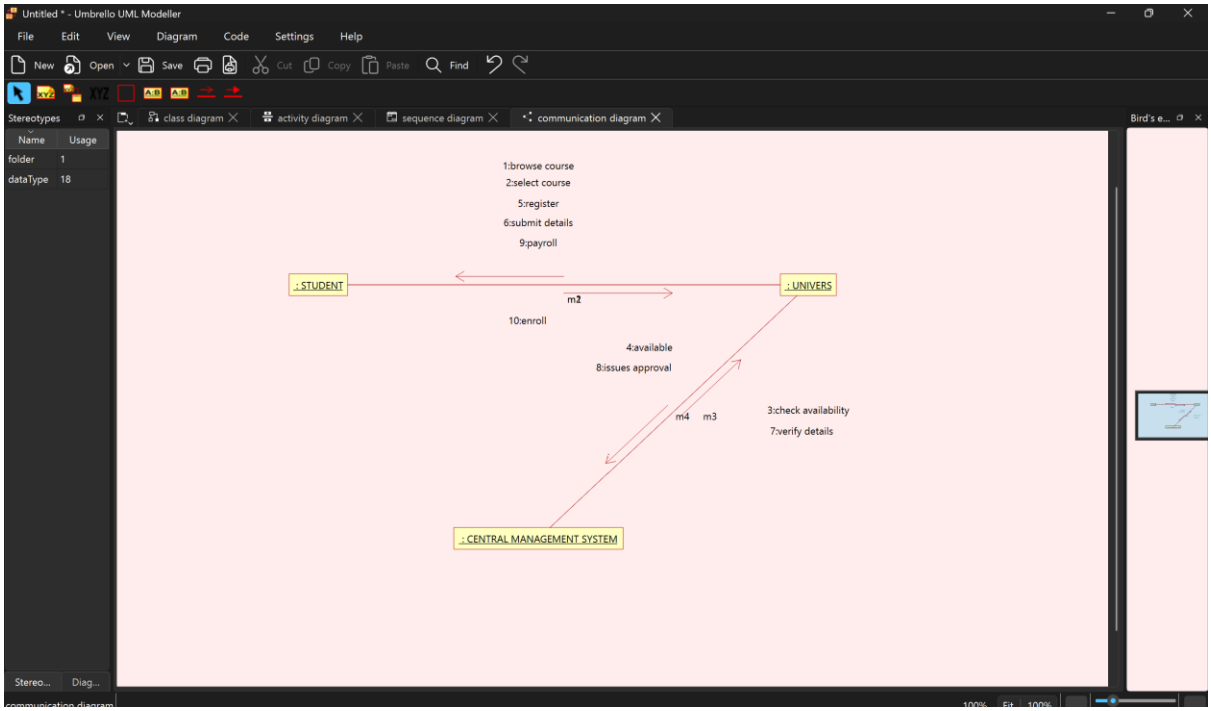
ACTIVITY DIAGRAM:



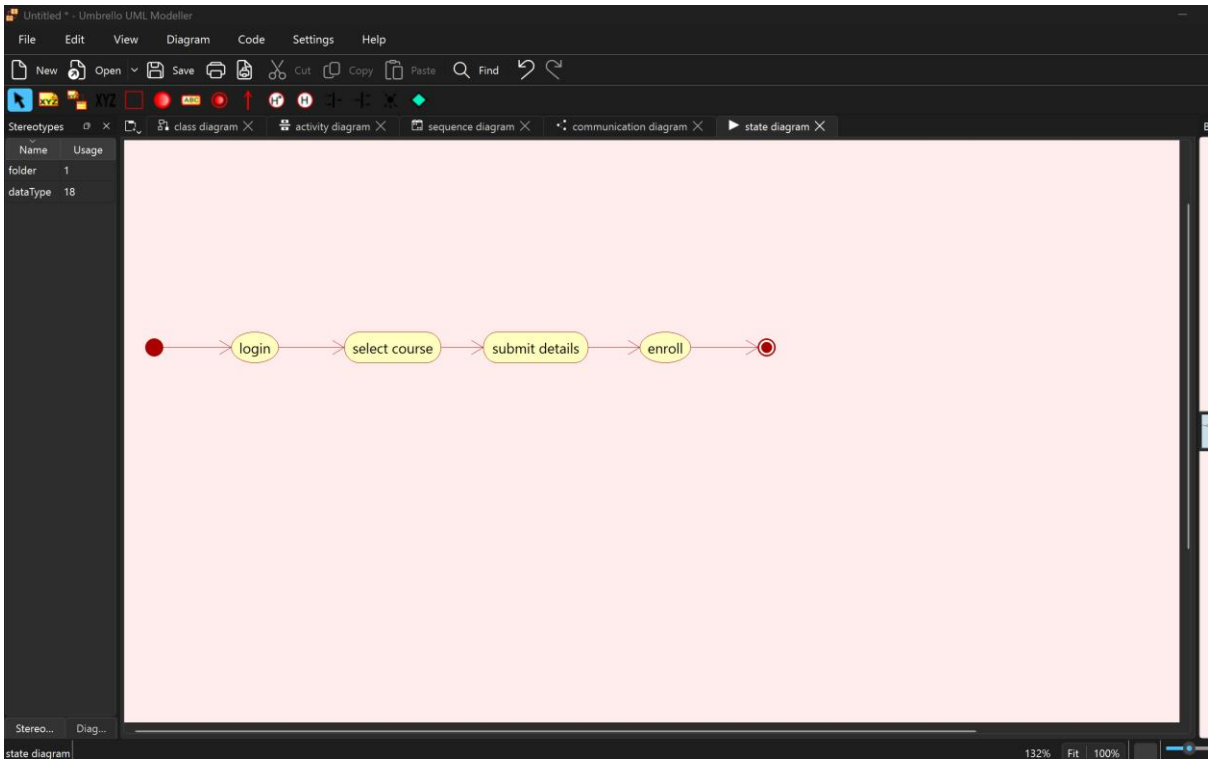
SEQUENCE DIAGRAM:



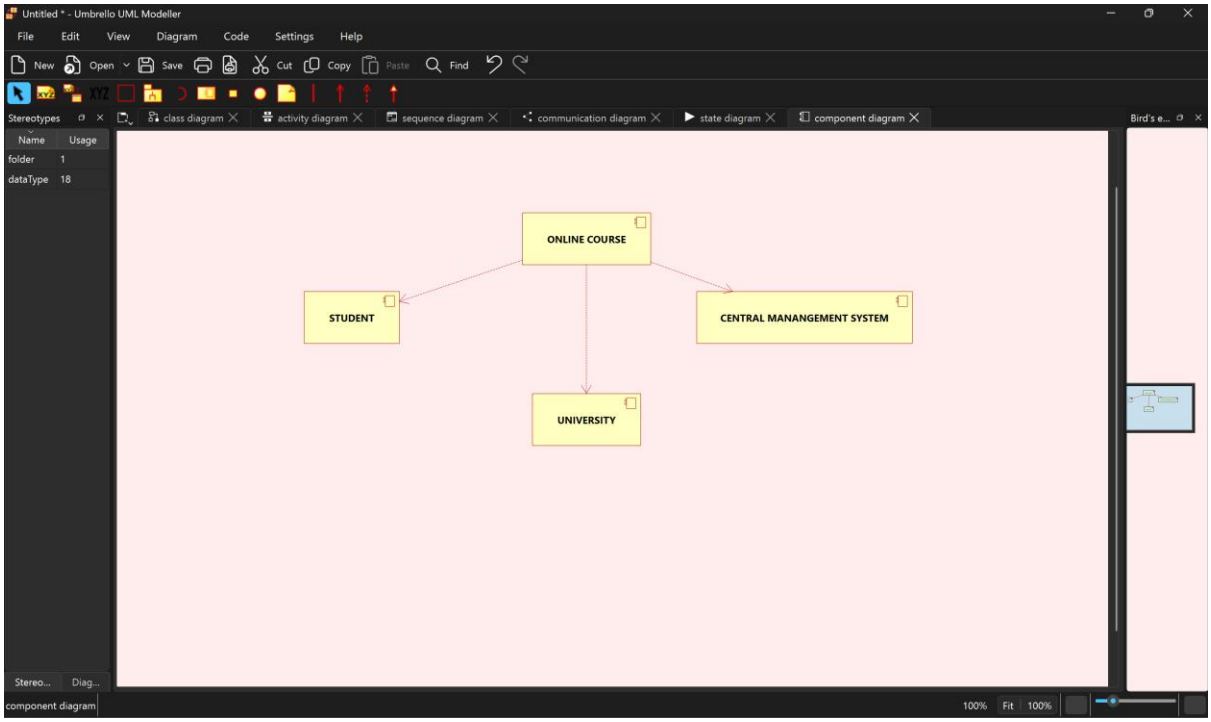
COLLABORATION DIAGRAM:



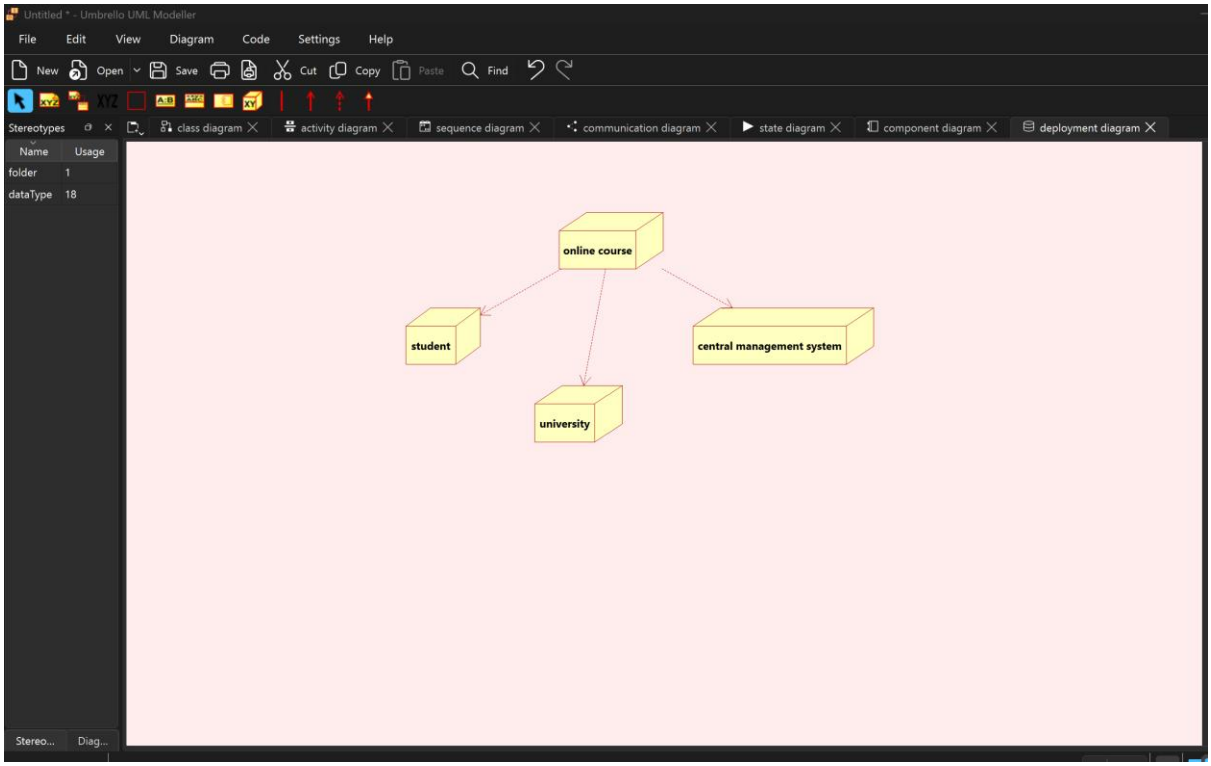
STATE CHART DIAGRAM:



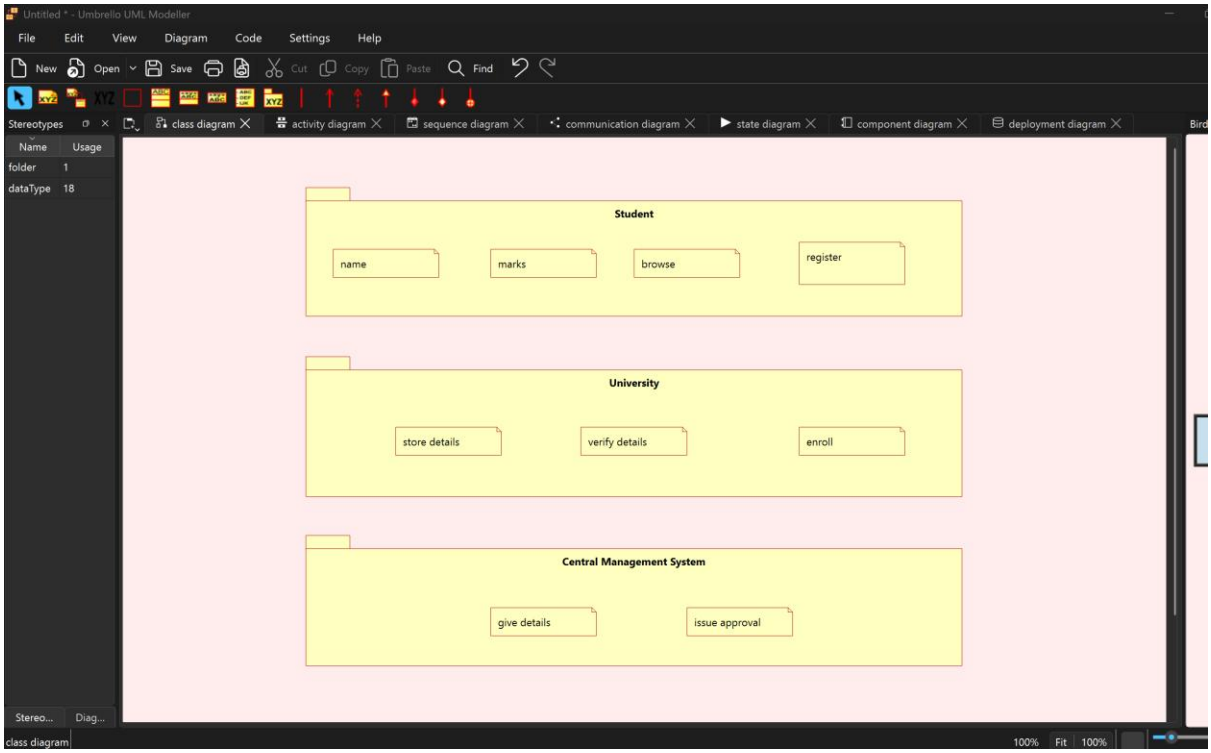
COMPONENT DIAGRAM:



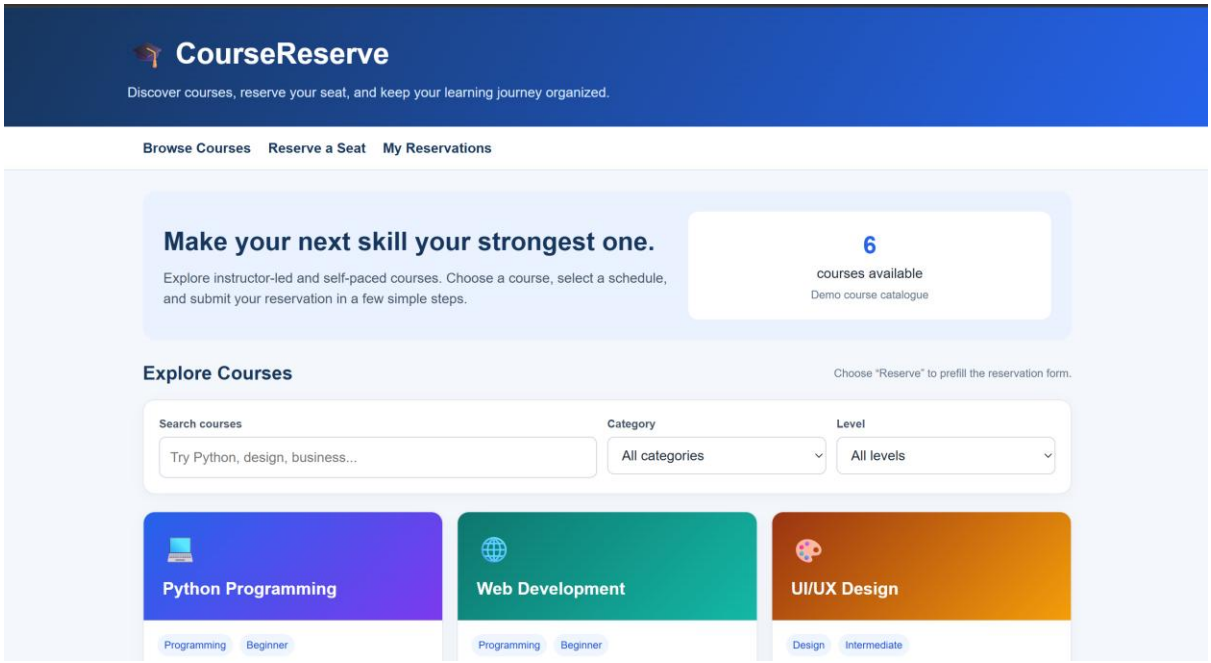
DEPLOYMENT DIAGRAM:

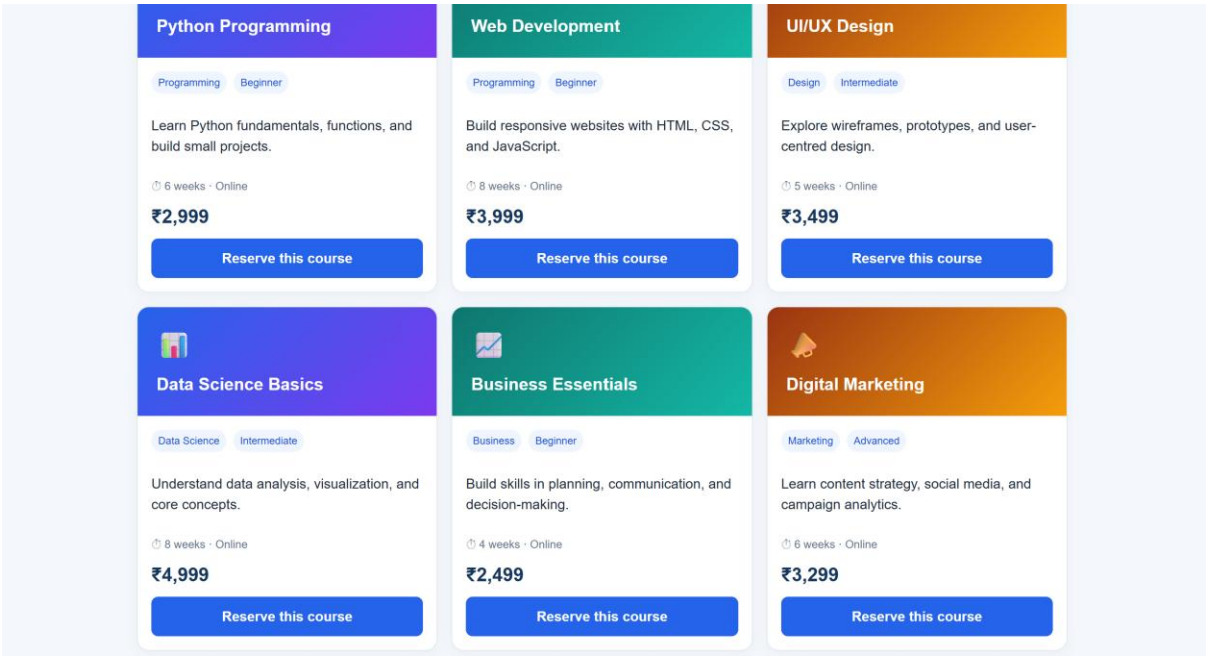


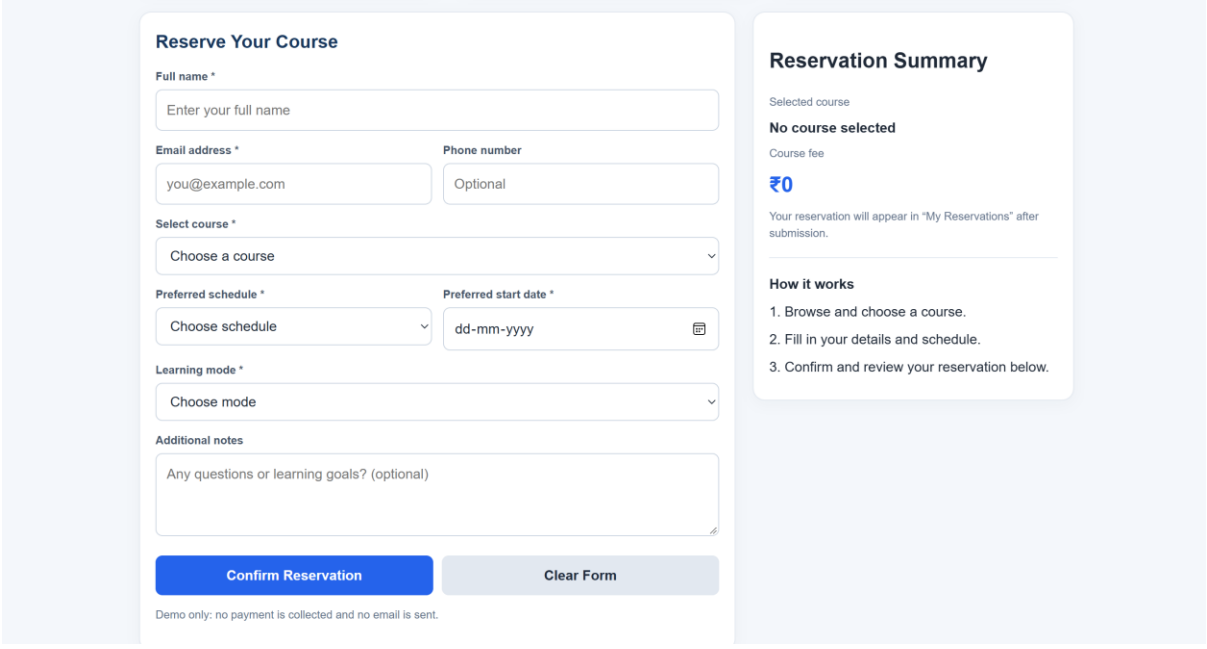
PACKAGE DIAGRAM

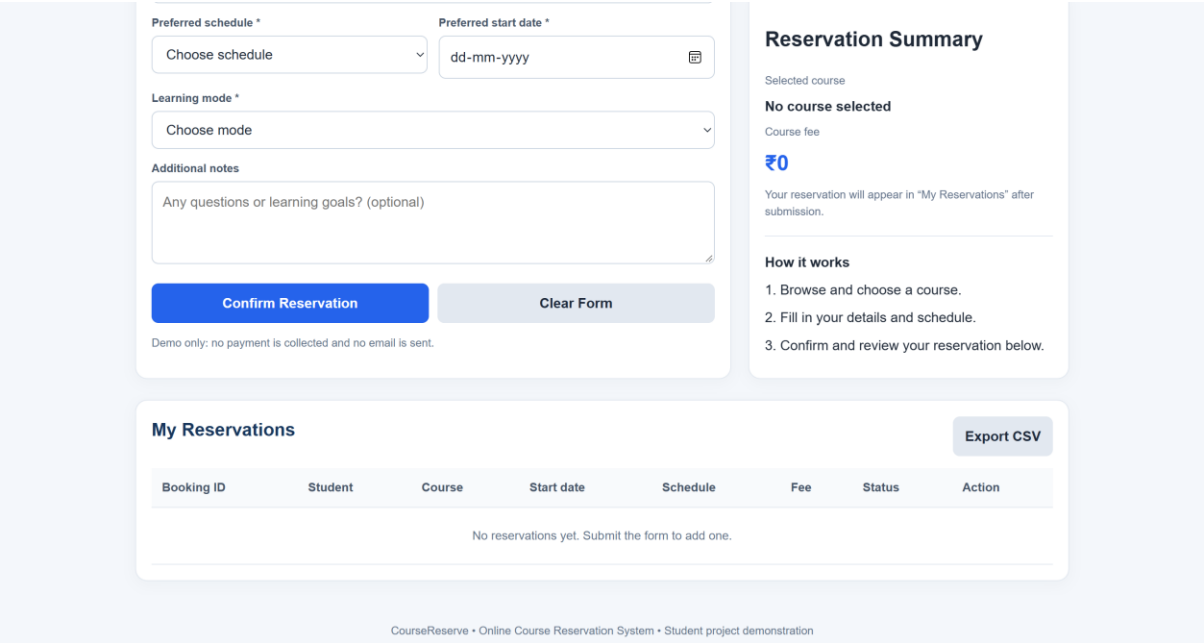


WEBPAGE FOR ONLINE COURSE RESERVATION SYSTEM




Python Programming
Programming
Beginner
Learn Python fundamentals, functions, and build small projects.
6 weeks · Online
₹2,999
Reserve this course
Web Development
Programming
Beginner
Build responsive websites with HTML, CSS, and JavaScript.
8 weeks · Online
₹3,999
Reserve this course
UI/UX Design
Design
Intermediate
Explore wireframes, prototypes, and user-centred design.
5 weeks · Online
₹3,499
Reserve this course
Data Science Basics
Data Science
Intermediate
Understand data analysis, visualization, and core concepts.
8 weeks · Online
₹4,999
Reserve this course
Business Essentials
Business
Beginner
Build skills in planning, communication, and decision-making.
4 weeks · Online
₹2,499
Reserve this course
Digital Marketing
Marketing
Advanced
Learn content strategy, social media, and campaign analytics.
6 weeks · Online
₹3,299
Reserve this course


Reserve Your Course
Full name *
Enter your full name
Email address *
you@example.com
Phone number
Optional
Select course *
Choose a course
Preferred schedule *
Choose schedule
Preferred start date *
dd-mm-yyyy
Learning mode *
Choose mode
Additional notes
Any questions or learning goals? (optional)
Confirm Reservation
Clear Form
Demo only: no payment is collected and no email is sent.
Reservation Summary
Selected course
No course selected
Course fee
₹0
Your reservation will appear in "My Reservations" after submission.
How it works
1. Browse and choose a course.
2. Fill in your details and schedule.
3. Confirm and review your reservation below.



HTML CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Online Course Reservation System</title>

<style>
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
```

```
  background: #f4f7fb;
  color: #1f2937;
}

header {
  background: linear-gradient(120deg, #17365d, #2563eb);
  color: white;
  padding: 25px;
  text-align: center;
}

header h1 {
  margin: 0;
}

nav {
  background: white;
  padding: 14px;
  text-align: center;
  border-bottom: 1px solid #ddd;
}

nav a {
  text-decoration: none;
  color: #17365d;
  font-weight: bold;
  margin: 0 12px;
}

main {
  max-width: 1100px;
```

```
    margin: auto;
    padding: 20px;
}

.hero {
    background: #eaf2ff;
    padding: 25px;
    border-radius: 12px;
    margin-bottom: 25px;
}

.hero h2 {
    color: #17365d;
}

.filters {
    display: grid;
    grid-template-columns: 2fr 1fr 1fr;
    gap: 12px;
    margin-bottom: 20px;
}

.course-grid {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 18px;
}

.course-card, .panel, .summary {
    background: white;
    border: 1px solid #e2e8f0;
```

```
    border-radius: 12px;
    padding: 18px;
    box-shadow: 0 3px 12px #00000008;
}

.course-card h3 {
    color: #17365d;
}

.course-card p {
    color: #64748b;
    font-size: 14px;
}

.tag {
    display: inline-block;
    background: #eff6ff;
    color: #1d4ed8;
    padding: 4px 8px;
    border-radius: 20px;
    font-size: 12px;
    margin-right: 5px;
}

.price {
    font-size: 22px;
    color: #17365d;
    font-weight: bold;
    margin: 12px 0;
}
```

```
input, select, textarea, button {
  width: 100%;
  padding: 11px;
  margin: 5px 0;
  border: 1px solid #cbd5e1;
  border-radius: 7px;
  font: inherit;
}

button {
  background: #2563eb;
  color: white;
  border: none;
  cursor: pointer;
  font-weight: bold;
}

button:hover {
  background: #17467f;
}

button.secondary {
  background: #64748b;
}

button.danger {
  background: #b91c1c;
}

.content-grid {
  display: grid;
```

```
    grid-template-columns: 1.4fr 0.8fr;
    gap: 20px;
    margin-top: 25px;
    align-items: start;
}

.panel h2, .summary h2 {
    color: #17365d;
}

.form-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 10px;
}

.full {
    grid-column: 1 / -1;
}

label {
    display: block;
    font-weight: bold;
    font-size: 13px;
    margin-top: 8px;
}

.summary .total {
    color: #2563eb;
    font-size: 25px;
    font-weight: bold;
```

```
}

.table-wrap {
  overflow-x: auto;
}

table {
  width: 100%;
  min-width: 700px;
  border-collapse: collapse;
  margin-top: 12px;
}

th, td {
  text-align: left;
  padding: 11px;
  border-bottom: 1px solid #e2e8f0;
  font-size: 14px;
}

th {
  background: #edf3fa;
}

.status {
  background: #dcfce7;
  color: #166534;
  padding: 4px 8px;
  border-radius: 15px;
  font-size: 12px;
}
```

```
.message {
  background: #dcfce7;
  color: #166534;
  padding: 12px;
  border-radius: 8px;
  display: none;
}

.small {
  color: #64748b;
  font-size: 13px;
}

footer {
  text-align: center;
  padding: 22px;
  color: #64748b;
}

@media (max-width: 800px) {
  .course-grid {
    grid-template-columns: repeat(2, 1fr);
  }

  .content-grid {
    grid-template-columns: 1fr;
  }
}

@media (max-width: 550px) {
```

```
  .course-grid, .filters, .form-grid {
    grid-template-columns: 1fr;
  }

  .full {
    grid-column: auto;
  }

  nav a {
    display: inline-block;
    margin: 6px;
  }
}
</style>
</head>

<body>

<header>
  <h1>🎓 CourseReserve</h1>
  <p>Online Course Reservation System</p>
  <p>Discover courses, reserve your seat, and learn new skills.</p>
</header>

<nav>
  <a href="#courses">Browse Courses</a>
  <a href="#reserve">Reserve a Course</a>
  <a href="#bookings">My Reservations</a>
</nav>

<main>
```

```
<section class="hero">
  <h2>Start Your Learning Journey!</h2>
  <p>Explore online courses, choose a suitable schedule, and reserve your seat easily.</p>
  <strong id="courseCount">6 courses available</strong>
</section>

<section id="courses">
  <h2>Explore Courses</h2>

  <div class="filters">
    <div>
      <label for="search">Search courses</label>
      <input id="search" placeholder="Search by course name">
    </div>

    <div>
      <label for="category">Category</label>
      <select id="category">
        <option value="">All Categories</option>
        <option>Programming</option>
        <option>Design</option>
        <option>Data Science</option>
        <option>Business</option>
        <option>Marketing</option>
      </select>
    </div>

    <div>
      <label for="level">Level</label>
```

```
        <select id="level">
          <option value="">All Levels</option>
          <option>Beginner</option>
          <option>Intermediate</option>
          <option>Advanced</option>
        </select>
      </div>
    </div>

    <div class="course-grid" id="courseGrid"></div>
</section>

<section class="content-grid" id="reserve">

    <div class="panel">
      <h2>Reserve Your Course</h2>

      <div class="message" id="message" role="status"></div>

      <form id="reservationForm">
        <div class="form-grid">

          <div class="full">
            <label for="studentName">Full Name *</label>
            <input id="studentName" required
                placeholder="Enter your full name">
          </div>

          <div>
            <label for="email">Email Address *</label>
            <input id="email" type="email" required
```

```
        placeholder="you@example.com">
</div>

<div>
  <label for="phone">Phone Number</label>
  <input id="phone" type="tel"
        placeholder="Enter phone number">
</div>

<div class="full">
  <label for="courseSelect">Select Course *</label>
  <select id="courseSelect" required>
    <option value="">Choose a course</option>
  </select>
</div>

<div>
  <label for="schedule">Preferred Schedule *</label>
  <select id="schedule" required>
    <option value="">Choose schedule</option>
    <option>Weekday Mornings</option>
    <option>Weekday Evenings</option>
    <option>Weekend Mornings</option>
    <option>Self-paced</option>
  </select>
</div>

<div>
  <label for="startDate">Start Date *</label>
  <input id="startDate" type="date" required>
</div>
```

```
        <div class="full">
          <label for="mode">Learning Mode *</label>
          <select id="mode" required>
            <option value="">Choose mode</option>
            <option>Online Live Class</option>
            <option>Self-paced Online</option>
          </select>
        </div>

        <div class="full">
          <label for="notes">Additional Notes</label>
          <textarea id="notes" rows="3"
                placeholder="Your learning goals (optional)"></textarea>
        </div>
      </div>

      <button type="submit">Confirm Reservation</button>
      <button type="reset" class="secondary">Clear Form</button>
    </form>

    <p class="small">
      Demo project only. No payment or email is processed.
    </p>
  </div>

  <aside class="summary">
    <h2>Reservation Summary</h2>
    <p class="small">Selected Course</p>
    <p id="summaryCourse">No course selected</p>
```

```
      <p class="small">Course Fee</p>
      <p class="total" id="summaryPrice">₹0</p>

      <hr>

      <h3>How It Works</h3>
      <p>1. Choose a course.</p>
      <p>2. Enter your details.</p>
      <p>3. Select a schedule and date.</p>
      <p>4. Confirm your reservation.</p>
    </aside>

</section>

<section class="panel" id="bookings" style="margin-top:25px">
    <h2>My Reservations</h2>

    <button id="exportBtn" style="max-width:180px">
      Export Reservations CSV
    </button>

    <div class="table-wrap">
      <table>
        <thead>
          <tr>
            <th>Booking ID</th>
            <th>Student</th>
            <th>Course</th>
            <th>Start Date</th>
            <th>Schedule</th>
            <th>Fee</th>
```

```
            <th>Status</th>
            <th>Action</th>
          </tr>
        </thead>
        <tbody id="bookingTable">
          <tr>
            <td colspan="8">No reservations yet.</td>
          </tr>
        </tbody>
      </table>
    </div>
</section>

</main>

<footer>
  CourseReserve | Online Course Reservation System
  <br>College Project Demonstration
</footer>

<script>
const courses = [
  {
    id: "python",
    name: "Python Programming",
    category: "Programming",
    level: "Beginner",
    duration: "6 weeks",
    fee: 2999,
    icon: "💻",
    description: "Learn Python basics, functions, and programming."
```

```
},
{
  id: "web",
  name: "Web Development",
  category: "Programming",
  level: "Beginner",
  duration: "8 weeks",
  fee: 3999,
  icon: "🌐",
  description: "Build websites using HTML, CSS, and JavaScript."
},
{
  id: "design",
  name: "UI/UX Design",
  category: "Design",
  level: "Intermediate",
  duration: "5 weeks",
  fee: 3499,
  icon: "🎨",
  description: "Learn wireframes, prototypes, and user experience."
},
{
  id: "data",
  name: "Data Science Basics",
  category: "Data Science",
  level: "Intermediate",
  duration: "8 weeks",
  fee: 4999,
  icon: "📊",
  description: "Explore data analysis and visualization."
```

```
  },
  {
    id: "business",
    name: "Business Essentials",
    category: "Business",
    level: "Beginner",
    duration: "4 weeks",
    fee: 2499,
    icon: "📈",
    description: "Improve business planning and communication skills."
  },
  {
    id: "marketing",
    name: "Digital Marketing",
    category: "Marketing",
    level: "Advanced",
    duration: "6 weeks",
    fee: 3299,
    icon: "📣",
    description: "Learn digital campaigns and marketing strategy."
  }
];

let reservations = [];

const $ = id => document.getElementById(id);

function escapeHTML(value) {
  return String(value).replace(/[&<>"']/g, function(c) {
    return {
      "&": "&amp;",
```

```
        "<": "&lt;",
        ">": "&gt;",
        '"': "&quot;",
        "'": "&#39;"
      }[c];
    });
}

function money(amount) {
  return "₹" + Number(amount).toLocaleString("en-IN");
}

function renderCourses() {
  const query = $("search").value.toLowerCase().trim();
  const category = $("category").value;
  const level = $("level").value;

  const filtered = courses.filter(course => {
    const matchesSearch =
      (course.name + " " + course.description)
      .toLowerCase().includes(query);

    return matchesSearch &&
      (!category || course.category === category) &&
      (!level || course.level === level);
  });

  $("courseCount").textContent =
    filtered.length + " courses available";

  $("courseGrid").innerHTML = filtered.length
```

```
      ? filtered.map(course => `
        <article class="course-card">
          <div style="font-size:30px">${course.icon}</div>
          <h3>${escapeHTML(course.name)}</h3>
          <span class="tag">${escapeHTML(course.category)}</span>
          <span class="tag">${escapeHTML(course.level)}</span>
          <p>${escapeHTML(course.description)}</p>
          <p><strong>Duration:</strong> ${escapeHTML(course.duration)}</p>
          <div class="price">${money(course.fee)}</div>
          <button type="button"
            onclick="chooseCourse('${course.id}')">
            Reserve This Course
          </button>
        </article>
      `).join("")
      : "<p>No courses match your search.</p>";
}

function fillCourseOptions() {
  $("courseSelect").innerHTML =
    '<option value="">Choose a course</option>' +
    courses.map(course =>
      `<option value="${course.id}">
        ${escapeHTML(course.name)} - ${money(course.fee)}
      </option>`
    ).join("");
}

function chooseCourse(id) {
  $("courseSelect").value = id;
  updateSummary();
```

```
  $("reserve").scrollIntoView({behavior: "smooth"});
}

function updateSummary() {
  const course = courses.find(
    item => item.id === $("courseSelect").value
  );

  $("summaryCourse").textContent =
    course ? course.name : "No course selected";

  $("summaryPrice").textContent =
    course ? money(course.fee) : "₹0";
}

function showMessage(text) {
  $("message").textContent = text;
  $("message").style.display = "block";
}

function renderReservations() {
  if (reservations.length === 0) {
    $("bookingTable").innerHTML =
      '<tr><td colspan="8">No reservations yet.</td></tr>';
    return;
  }

  $("bookingTable").innerHTML = reservations.map(booking => `
    <tr>
      <td>${escapeHTML(booking.id)}</td>
      <td>${escapeHTML(booking.name)}</td>
```

```
        <td>${escapeHTML(booking.courseName)}</td>
        <td>${escapeHTML(booking.date)}</td>
        <td>${escapeHTML(booking.schedule)}</td>
        <td>${money(booking.fee)}</td>
        <td><span class="status">${booking.status}</span></td>
        <td>
          <button class="danger" type="button"
            onclick="cancelReservation('${booking.id}')">
            Cancel
          </button>
        </td>
      </tr>
    `).join("");
}

$("reservationForm").addEventListener("submit", function(event) {
  event.preventDefault();

  const course = courses.find(
    item => item.id === $("courseSelect").value
  );

  if (!course) {
    showMessage("Please select a course.");
    return;
  }

  const date = $("startDate").value;
  const today = new Date();
  today.setHours(0, 0, 0, 0);
```

```
  if (new Date(date + "T00:00:00") < today) {
    showMessage("Please select today or a future date.");
    return;
  }

  const booking = {
    id: "CR" + Date.now().toString().slice(-7),
    name: $("studentName").value.trim(),
    email: $("email").value.trim(),
    phone: $("phone").value.trim(),
    courseName: course.name,
    fee: course.fee,
    date: date,
    schedule: $("schedule").value,
    mode: $("mode").value,
    notes: $("notes").value.trim(),
    status: "Reserved"
  };

  reservations.unshift(booking);
  renderReservations();

  showMessage(
    "Reservation confirmed! Your booking ID is " + booking.id
  );

  this.reset();
  updateSummary();
});

$("reservationForm").addEventListener("reset", function() {
```

```
  setTimeout(updateSummary, 0);
});

function cancelReservation(id) {
  if (confirm("Do you want to cancel this reservation?")) {
    reservations = reservations.filter(
      booking => booking.id !== id
    );
    renderReservations();
    showMessage("Reservation cancelled.");
  }
}

function csvCell(value) {
  return '"' + String(value).replace(/"/g, '""') + '"';
}

$("exportBtn").addEventListener("click", function() {
  if (reservations.length === 0) {
    alert("There are no reservations to export.");
    return;
  }

  const headers = [
    "Booking ID", "Student", "Email", "Phone",
    "Course", "Start Date", "Schedule",
    "Learning Mode", "Fee", "Status", "Notes"
  ];

  const rows = reservations.map(booking => [
    booking.id, booking.name, booking.email,
```

```
      booking.phone, booking.courseName, booking.date,
      booking.schedule, booking.mode, booking.fee,
      booking.status, booking.notes
    ]);

    const csv = [headers, ...rows]
      .map(row => row.map(csvCell).join(","))
      .join("\r\n");

    const blob = new Blob(["\uFEFF" + csv], {
      type: "text/csv;charset=utf-8;"
    });

    const url = URL.createObjectURL(blob);
    const link = document.createElement("a");
    link.href = url;
    link.download = "course_reservations.csv";
    link.click();
    URL.revokeObjectURL(url);
});

["search", "category", "level"].forEach(id => {
    $(id).addEventListener(
      id === "search" ? "input" : "change",
      renderCourses
    );
});

$("courseSelect").addEventListener("change", updateSummary);

const now = new Date();
```

```
const localToday = new Date(
   now.getTime() - now.getTimezoneOffset() * 60000
).toISOString().slice(0, 10);

$("startDate").min = localToday;

fillCourseOptions();
renderCourses();
renderReservations();
updateSummary();
</script>

</body>
</html>
```